Setting up Variational Eigenproblems for Colbrook's Algorithm

August 19, 2026

This note clarifies our confusion about how to set up variational eigenproblems for Colbrook's algorithm for computing spectra and pseudospectra.

1 The setting of Colbrook

We exclusively consider the folding strategy. Folding starts from a closed, densely defined A on a Hilbert space $\mathcal{H}$. Since we want its spectrum, A must be an *endomorphism*, $A\colon \mathrm{D}(A) \subset \mathcal{H} \to \mathcal{H}$, the classical instance being $\Delta\colon H^2(\Omega) \cap H_0^1(\Omega) \to L^2(\Omega)$, i.e. $\mathcal{H} = L^2(\Omega)$, and $\mathrm{D}(A) = H^2(\Omega) \cap H_0^1(\Omega) \subset \mathcal{H}$. We use * for the Hilbert adjoint of a densely defined endomorphism on $\mathcal{H}$, and $'$ for the Banach dual operator: if $B : X \to Y$ is bounded, then $B' : Y^* \to X^*$. Under suitable conditions on $\mathrm{D}(A)$ and $\mathrm{D}(A^*)$ one considers

$$\gamma(z, A) := \min\bigl\{\sigma_{\inf}(A - zI),\ \sigma_{\inf}\bigl((A - zI)^*\bigr)\bigr\} = \bigl\|(A - zI)^{-1}\bigr\|^{-1}, \tag{1}$$

where the injection moduli come from the quadratic forms of the resolvent and its adjoint,

$$\sigma_{\inf}(A - zI)^2 = \inf_{f \in \mathrm{D}(A)\setminus\{0\}} \frac{\|(A - zI)f\|_{\mathcal{H}}^2}{\|f\|_{\mathcal{H}}^2}, \qquad \sigma_{\inf}\bigl((A - zI)^*\bigr)^2 = \inf_{f \in \mathrm{D}(A^*)\setminus\{0\}} \frac{\|(A - zI)^* f\|_{\mathcal{H}}^2}{\|f\|_{\mathcal{H}}^2}. \tag{2}$$

This reduces the computation of pseudospectra to the eigenvalues of the self-adjoint operator

$$(A - zI)^*(A - zI). \tag{3}$$

2 The standard Dirichlet Laplace eigenproblem

The standard variational setup for the Dirichlet Laplace eigenproblem does not directly give us the endomorphism A above. Instead, it involves two different operators acting on different spaces.

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain. We define the trial space $V = H_0^1(\Omega)$ and the pivot space $\mathcal{H} = L^2(\Omega)$, so that we have the Gelfand triple $V \subset \mathcal{H} \cong \mathcal{H}^* \subset V^*$.

First, we define the *weak Laplacian* $L\colon H_0^1(\Omega) \to H^{-1}(\Omega)$. This operator is defined via the Dirichlet form:

$$\langle Lu, v\rangle_{V^*,V} = (\nabla u, \nabla v)_{L^2(\Omega)} \quad \text{for all } u, v \in H_0^1(\Omega). \tag{4}$$

Second, we define the *mass operator* $M\colon L^2(\Omega) \to V^*$, which maps an L^2 function u to the linear functional that computes the L^2-inner product with u:

$$\langle Mu, v\rangle_{V^*,V} = (u, v)_{L^2(\Omega)} \quad \text{for all } v \in H_0^1(\Omega). \tag{5}$$

Note that the standard setup typically writes this simply as the L^2 inner product, but interpreting M as an operator into V^* is the precise functional analytic setting.

The standard variational eigenproblem is then to find $\lambda \in \mathbb{C}$ and $u \in H_0^1(\Omega) \setminus \{0\}$ such that

$$Lu = \lambda Mu \quad \text{in } V^*. \tag{6}$$

These are the ingredients we have at our disposal to define the densely defined endomorphic operator A required for Colbrook's folding algorithm.

3 Different approaches for defining the endomorphism

We explore different ways to combine L and M into an endomorphism.

The solution operator approach. One can define the solution operator T mapping $L^2(\Omega)$ right-hand sides to solutions in $L^2(\Omega)$ (this is the approach taken by Boffi). Formally, we define

$$T = L^{-1}M \colon L^2(\Omega) \to L^2(\Omega). \tag{7}$$

More precisely, for $f \in L^2(\Omega)$, $Tf = u \in H_0^1(\Omega) \subset L^2(\Omega)$ is the unique solution to $Lu = Mf$. Because of the compact embedding $H_0^1(\Omega) \subset\subset L^2(\Omega)$, T is a *compact* operator on $L^2(\Omega)$. Its non-zero eigenvalues are the reciprocals of the eigenvalues of the original problem ($Tu = \lambda^{-1}u$). However, since T is compact, its spectrum is fundamentally different from an unbounded differential operator (it only has essential spectrum at 0). It is a different beast and not what Colbrook's algorithm is designed to study when analyzing spectral pollution and essential spectra.

Applying the Riesz map. Could we apply the Riesz map of $H_0^1(\Omega)$ to bridge the gap? The Riesz map $R_V \colon V \to V^*$ is defined by $\langle R_V u, v\rangle = (u, v)_V$. If we use the standard H_0^1 inner product $(u, v)_V = (\nabla u, \nabla v)_{L^2}$, then the Riesz map R_V coincides exactly with the weak Laplacian L. Therefore, applying the inverse Riesz map R_V^{-1} to L just means applying L^{-1} to L, which gives the identity on $H_0^1(\Omega)$. On the other hand, applying it to the right-hand side means applying $L^{-1}M$, which returns us to the compact solution operator $T = L^{-1}M$ discussed above.

The $M^{-1}L$ approach. Does $M^{-1}L$ work to define an unbounded endomorphism A? Let us check if the spaces match up. The weak Laplacian L maps $H_0^1(\Omega) \to H^{-1}(\Omega)$. However, M^{-1} is only defined on the range of M, which comprises exactly those functionals in $H^{-1}(\Omega)$ that can be represented by L^2 functions. Thus, $A = M^{-1}L$ is only well-defined on the subspace of $u \in H_0^1(\Omega)$ for which $Lu \in \text{range}(M) \subset H^{-1}(\Omega)$. Identifying range$(M)$ with $L^2(\Omega)$, this domain is precisely $D(A) = \{u \in H_0^1(\Omega) : \Delta u \in L^2(\Omega)\}$, which is the graph space from Section 1. So $A = M^{-1}L$ does define the correct unbounded endomorphism $A \colon D(A) \subset L^2(\Omega) \to L^2(\Omega)$. Of course, the computational problem with this approach is that standard H^1-conforming finite elements do not lie in $D(A)$, making it difficult to discretise this operator directly for the folding algorithm without C^1 elements, but this is a separate (and fixable) difficulty.

However, note that M is not always invertible (e.g. an eigenproblem for the incompressible Stokes equations, with the right-hand side only depending on the velocity), so this is not a general solution.

The variational viewpoint. We have considered the operator $M^{-1}L$ where L is the "weak" Laplacian and M is the L^2 Riesz map, i.e.

$$L : H_0^1(\Omega) \to H^{-1}(\Omega), \qquad M : L^2 \to (L^2)^*, \tag{8}$$

unfortunately as pointed out above on the all $H_0^1(\Omega)$ the operator $M^{-1}L$ is not well defined and we try to repair this via the endomorphism A with its own domain of definition which is precisely,

$$A : L^2 \to L^2, \qquad A = M^{-1}L_{|_D}, \qquad D = \{u \in H_0^1(\Omega) \ : \ \Delta u \in L^2(\Omega)\}. \tag{9}$$

Notice that unless our domain is convex and we can apply a Miranda–Talenti inequality we have that $D \neq H^2 \cap H_0^1$. This approach comes with various issues. First, it is not familiar to variational people. Second and more importantly the operator is unbounded and its solution operator does not actually fit the Babuška–Obsorn theory that we would like to use to have a priori error estimates. This because this theory relies on the pair (G, T) of a bounded linear operator $G : V \to V^*$ and its corresponding compact solution operator $T : \mathcal{H} \to B \subset \mathcal{H}$.

For this reason we would like propose an equivalent but different functional analytical viewpoint. It will have the same computations, but permit the application of Babuška–Osborn theory. It rests on the following calculations. Noting that

$$M^{-1}(L - zM) = M^{-1}L - zI = A - zI \tag{10}$$

and

$$L - zM = M\big(M^{-1}L - zI\big) = M(A - zI), \tag{11}$$

we set $C_z \coloneqq M^{-1}(L - zM) = A - zI : V \to \mathcal{H}$, let $C_z' : \mathcal{H}^* \to V^*$ denote its dual operator, and let $J_{\mathcal{H}} : \mathcal{H} \to \mathcal{H}^{**}$ be the canonical injection. Since $L - zM = MC_z$ and the L^2 Riesz map satisfies $M'J_{\mathcal{H}} = M$, the variational folding operator is

$$\begin{aligned} G \coloneqq (L - zM)'J_{\mathcal{H}}M^{-1}(L - zM) &= (MC_z)'J_{\mathcal{H}}C_z \\ &= C_z'M'J_{\mathcal{H}}C_z \\ &= C_z'MC_z, \end{aligned} \tag{12}$$

where we used the fact that the inverse of the canonical bidual identification is $J_{\mathcal{H}}$ it self. Recall that here the primes denote Banach dual operators, not the Hilbert adjoint of the unbounded operator A. The previous equation tells us something very interesting: both the folding of the endomorphic A and the operator $G : V \to V^*$ are associated to the same bilinear form (that we can assemble in Firedrake!), i.e.

$$\begin{aligned} a_z(u, v) &\coloneqq \big\langle (L - zM)'J_{\mathcal{H}}M^{-1}(L - zM)u, v\big\rangle_{(V^*,V)} \\ &= \langle C_z'MC_zu, v\rangle_{(V^*,V)} \\ &= \big(C_zu, C_zv\big)_{\mathcal{H}} = \big((A - zI)u, (A - zI)v\big)_{\mathcal{H}}. \end{aligned} \tag{13}$$

Let us now take a closer look at the operator $G : V \to V^*$. This operator is perfectly well-defined, using the canonical injection into the double dual, if $(L - zM)$ maps from V to $\mathcal{H}^*$:

$$V \overset{L-zM}{\longrightarrow} \mathcal{H}^* \overset{M^{-1}}{\longrightarrow} \mathcal{H} \overset{J_{\mathcal{H}}}{\longrightarrow} \mathcal{H}^{**} \overset{(L-zM)'}{\longrightarrow} V^*. \tag{14}$$

This setting works for the following scenarios:

1. Dirac folding, $V = H^1(\Omega) \times H_0^1(\Omega)$ and $\mathcal{H} = L^2(\Omega)$;
2. First order Maxwell folding, $V = H_0(\mathrm{curl};\,\Omega) \cap H(\mathrm{div};\Omega) \times H_0(\mathrm{curl};\,\Omega) \cap H(\mathrm{div};\Omega)$ and $\mathcal{H} = L^2(\Omega)$;
3. Laplace folding, $V = H_0^1(\Omega) \cap H^2(\Omega)$ and $\mathcal{H} = L^2(\Omega)$, with Ω convex.

Notice that in all this cases we have $V \subset\subset \mathcal{H}$, and hence we can introduce (with no use of elliptic regularity!) the solution operator associated to $G : V \to V^*$, i.e.

$$T : \mathcal{H} \to \mathcal{H}, \qquad a(Tf, v) = (f, v)_H, \qquad \forall v \in V, \tag{15}$$

which in turns allow us to make use of the Babuška–Osborn theory as explained in Boffi's Acta Numerica article.

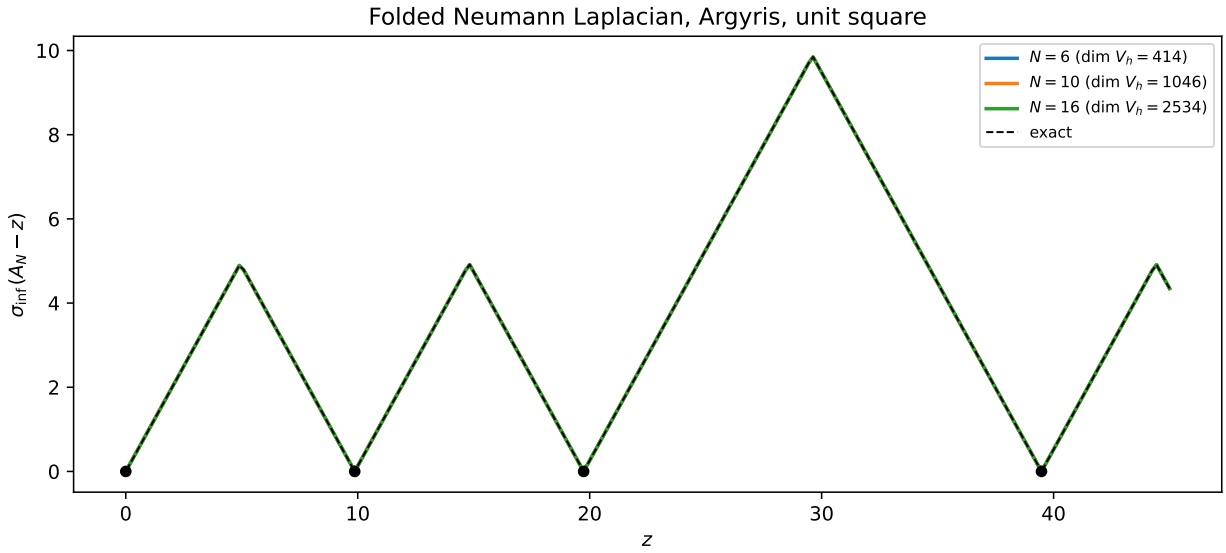


Figure 1: Unit square, which is convex, so $\mathrm{D}(A_N) = H^2(\Omega) \cap \{\partial_\nu u = 0\}$ and the Argyris space is conforming. $\sigma_{\inf}(A_N - z)$ reproduces $\mathrm{dist}(z, \sigma(A_N))$ over $z \in [0, 45]$ with maximum error $2.75 \cdot 10^{-4}$, $5.66 \cdot 10^{-6}$, $1.54 \cdot 10^{-7}$ for $N = 6, 10, 16$; the three curves are indistinguishable from the exact sawtooth at this scale.

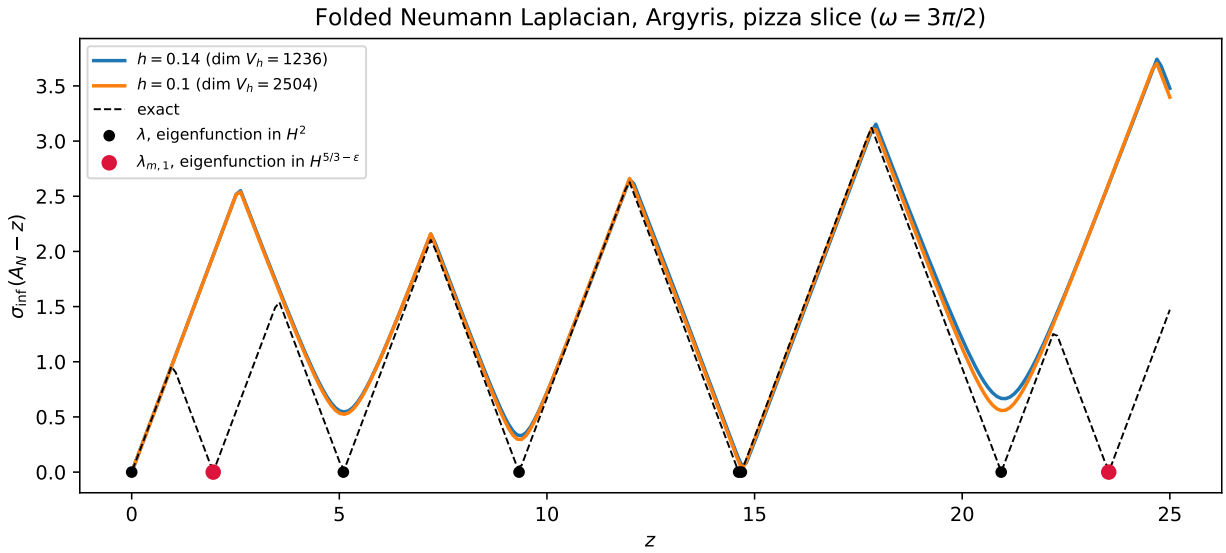


Figure 2: Pizza slice of aperture $\omega = \frac{3}{2}\pi$, where the convexity assumption in case 3 fails. The eigenfunctions $\Phi_{m,n} \sim \rho^{2n/3} \cos(\frac{2}{3} n\theta)$ with $n = 1$ lie in $H^{5/3-\epsilon}$ and *not* in H^2, and since $H^2 \cap \{\partial_\nu u = 0\}$ is a closed subspace of $\mathrm{D}(A_N)$ in the graph norm, no H^2-conforming space can reach them. The computed curves touch zero at every black marker and walk straight past the two red ones, $\lambda_{m,1} = (j'_{2/3,m})^2 = 1.9634119,\ 23.5286590$; the floor there is unmoved by refinement or by grading toward the re-entrant corner.

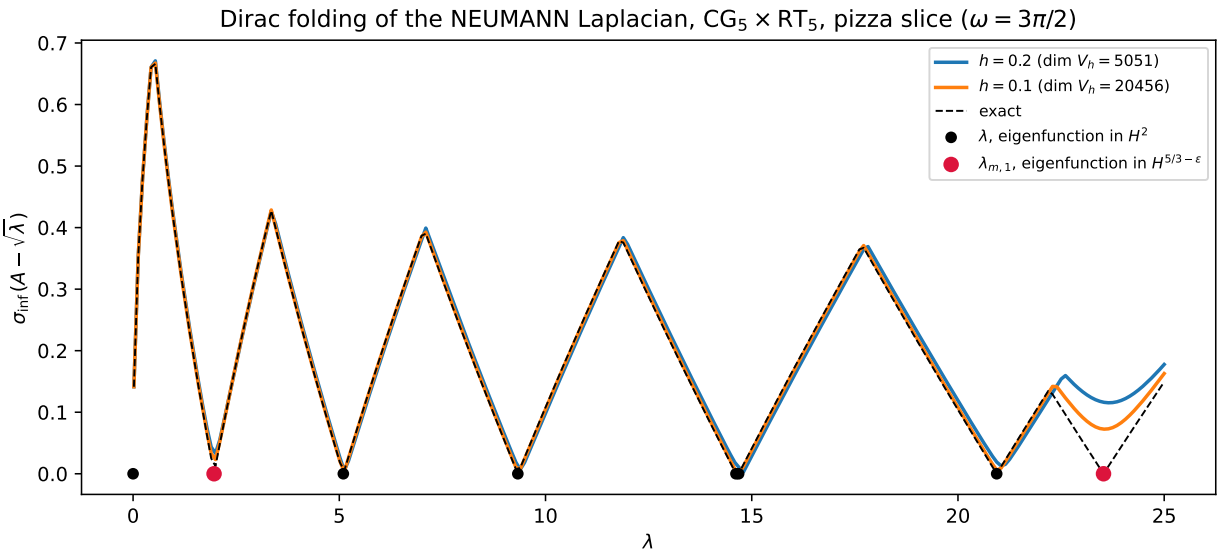


Figure 3: The same domain and the same operator as Figure 2, folded the other way: case 1 instead of case 3. Rather than A_N itself we fold the first-order $A(u, \sigma) = (-i \operatorname{div} \sigma, -i \nabla u)$, whose square is A_N once $\sigma \cdot n = 0$, that being exactly $\partial_\nu u = 0$ since $\sigma = -i\nabla u/\mu$. Now $\mathrm{D}(A) = H^1(\Omega) \times H_0(\mathrm{div}; \Omega)$ contains the singular eigenfunctions outright, $u \sim \rho^{2/3} \in H^1$, $\sigma \sim \rho^{-1/3} \in (L^2)^2$, $\operatorname{div} \sigma \propto u \in L^2$, so they are reachable, and the computed curves touch zero at the two red markers $\lambda_{m,1} = 1.9634,\ 23.5287$ that Figure 2 ignores. Elements are $\mathrm{CG}_5 \times \mathrm{RT}_5$, since in the Neumann case the essential condition sits on the flux.